

Natural Language Processing
Programing Task-2

Saggu Nagaraju

Task 1 – Customized Text Production Using Decoding Strategies

Sampling Strategies Tested:

We evaluated various decoding techniques using a custom decoder script implemented in `decoder.py`, applied on Alpaca-style instruction prompts. Configurations included:

- **Temperature Sampling:** 0.5, 0.7, 1.0
- **Top-k Filtering:** 10, 50
- **Top-p (Nucleus Sampling):** 0.5, 0.95
- **Hybrid Sampling:** Temperature=0.7, Top-k=50, Top-p=0.9
- **do_sample:** Enabled and disabled

Generation Quality Observations:

- **Greedy Decoding:** Predictable and repetitive, often reiterating the prompt
- **Temperature 0.5:** Produced overly cautious and less meaningful results
- **Top-k 50:** Generated more diverse and fluid outputs
- **Top-p 0.95:** Balanced coherence with creativity
- **Hybrid:** Showed the best overall performance across consistency and novelty

Strategy Comparison Summary

Sampling Strategy	Output Quality
Greedy	Repetitive / Low novelty
Temperature = 0.5	Conservative and fragile
Top-k = 50	Diverse, some artifacts
Top-p = 0.95	Good fluency and variation
Hybrid (0.7/50/0.9)	Best balance of coherence & creativity

Conclusion: Hybrid decoding provided the most reliable and human-like outputs. It is recommended for tasks requiring instruction adherence and expressive language generation.

Exercise 2: Task 2 – Fine-Tuning GPT-2 on Instruction Dataset

Fine-tuning Setup :

We fine-tuned the GPT-2 model on a subset of 10 Alpaca-style instruction-following pairs (8 for training, 2 for testing).

- **Epochs:** 1
- **Batch Size:** 4
- **Learning Rate:** 5e-5
- **Max Length:** 128 tokens
- **Loss:** Causal Language Modeling Loss
- **Evaluation Metrics:** Perplexity, Repetition Score, Lexical Diversity, Jaccard Similarity

Quantitative Results :

Metric	Value
Perplexity	277.4
Repetition Score	0.2801
Diversity Score	0.4888
Jaccard Similarity	0.5177

- High perplexity reflects the challenge of learning from a very small dataset.
- Repetition was moderate, and diversity reasonably acceptable.
- Jaccard similarity showed decent lexical overlap between model and expected outputs.

Qualitative Examples :

- **Prompt:** "Give three tips for staying healthy"
→ Model output included relevant tips but repeated some lines.

Conclusion

Fine-tuning improved task alignment and formatting behaviour but suffered from limited data and training. Despite high perplexity, generated texts were instruction-aligned and mostly coherent.